

ELL8299/COL8383/COL873: Trustworthy Large Language Models

Quiz 1 (Semester I, 2026-27)

Answer the questions in the spaces provided. No extra pages will be given.

Name: _____ Entry Number: _____

Total Marks: 20

Time: 60 Minutes

17/08/2026

1. **Language Modelling.** Modern language models rely on a foundational pipeline to process and generate text. Understanding their behavior requires examining how input text is segmented via tokenization, how architectures like multi-head attention capture contextual dependencies, and how metrics such as perplexity quantify the model's predictive uncertainty.

(a) Consider a vocabulary V containing K words. If a unigram language model assigns a uniform probability distribution $P(w) = 1/K$ to every word in V , what is the perplexity of any test sequence W of length M ? [2]

(b) Why is subword tokenization (e.g., BPE, WordPiece) preferred over pure word-level tokenization for LLMs? [2]

(c) Consider a multi-head attention layer with fixed embedding dimension d , equally divided across h heads. Would increasing h increase the total number of trainable parameters in the attention block? Explain why or why not. [1]

2. **Parameter-Efficient Fine-Tuning.** As pretrained language models scale to billions of parameters, full fine-tuning becomes increasingly expensive in terms of computation and memory. Parameter-Efficient Fine-Tuning (PEFT) reduces this cost by training only a small number of additional parameters. One widely used PEFT method is Low-Rank Adaptation (LoRA). Consider a weight matrix $W \in \mathbb{R}^{8192 \times 8192}$. In LoRA, the original matrix W is kept frozen, and its task-specific update is parameterized as $W' = W + BA$, where $A \in \mathbb{R}^{r \times 8192}$ and $B \in \mathbb{R}^{8192 \times r}$. Assume $r = 16$.

(a) How many parameters would a full fine-tuning update require for this matrix? [1]

(b) How many trainable parameters does LoRA introduce? [1]

(c) Approximately what fraction of the original parameters are trainable under LoRA? [1]

(d) Since BA is the update applied to W , what is the maximum possible rank of the update matrix BA ? How does this compare with the maximum possible rank of W ? [2]

3. **Memorization in LLMs.** The secret sharer setup, as discussed in class, evaluates unintended memorization by inserting random canaries $s[r]$ drawn from a randomness space $\mathcal{R}$. The exposure metric is defined as:

$$\text{exposure}_\theta(s[r]) = \log_2 |\mathcal{R}| - \log_2 \text{rank}_\theta(s[r])$$

Assume an adversary plants a canary with a fixed prefix followed by a random 6-character alphanumeric secret (using 26 lowercase letters and 10 digits). The target model with parameters θ is evaluated, and exactly 4,095 other candidate secrets from the randomness space $\mathcal{R}$ have a strictly lower perplexity than the true planted secret $s[r]$.

- (a) Calculate the value of $\text{exposure}_\theta(s[r])$. Show your intermediate calculations. *You may leave the final answer as an exact logarithmic expression if it cannot be simplified further.* [3]

- (b) Two canaries s_1 and s_2 are drawn from the same randomness space $\mathcal{R}$. Suppose the model ranks them as $\text{rank}_\theta(s_1) = 2^{10}$ and $\text{rank}_\theta(s_2) = 2^{14}$. Which of the following statements are **TRUE**? *Select all that apply. No partial credits.* [2]

- A. s_1 has 4 bits higher exposure than s_2 .
- B. s_2 has higher exposure because its rank is larger.
- C. The model assigns a better rank to s_1 than to s_2 .
- D. The exposure difference cannot be determined without knowing $|\mathcal{R}|$.

4. **Privacy through Re-identification.** Removing direct identifiers does not necessarily prevent re-identification, since combinations of other attributes may uniquely identify individuals. A dataset satisfies k -anonymity with respect to a set of quasi-identifiers if every combination of quasi-identifier values appearing in the dataset is shared by at least k records. The records are partitioned into equivalence classes according to their combinations of quasi-identifier values. A university releases the following supposedly anonymized dataset. Assume that Age, Gender, Department, and City are quasi-identifiers, while Academic Support/Award is a sensitive attribute.

Row	Age	Gender	Department	City	Academic Support/Award
1	28	F	CSE	Delhi	Industry Fellowship
2	34	F	CSE	Delhi	Best Thesis Award
3	31	F	CSE	Noida	Institute TAs
4	61	M	EE	Delhi	Research Assistantship
5	64	M	EE	Noida	Institute TAs
6	33	M	CSE	Delhi	Industry Fellowship

- (a) With respect to the quasi-identifiers, what is the value of k for the released dataset? [1]

- A. $k = 1$
- B. $k = 2$
- C. $k = 3$
- D. $k = 6$

- (b) Suppose Age is generalized into the intervals 20–39 and 60–79, and both Delhi and Noida are generalized to “NCR”. Gender and Department remain unchanged. Which row still belongs to an equivalence class of size 1? [1]

- A. Row 1
- B. Row 3
- C. Row 5
- D. Row 6

- (c) In addition to the transformations in (b), suppose Gender is suppressed. What is the resulting value of k with respect to the remaining quasi-identifiers? [1]

- A. $k = 1$
- B. $k = 2$
- C. $k = 3$
- D. $k = 4$

- (d) Based on the transformed dataset from (c), which of the following statements are **TRUE**? *Select all that apply.* [2]

- A. The transformed dataset satisfies 2-anonymity with respect to the remaining quasi-identifiers.
- B. Satisfying 2-anonymity guarantees that an attacker cannot infer the sensitive attribute of an individual with probability greater than $1/2$.
- C. Generalizing Delhi and Noida to “NCR” reduces the ability to distinguish individual records while retaining some geographic information.
- D. Removing the sensitive attribute from the original dataset would, by itself, increase its k -anonymity from $k = 1$ to at least $k = 2$.